

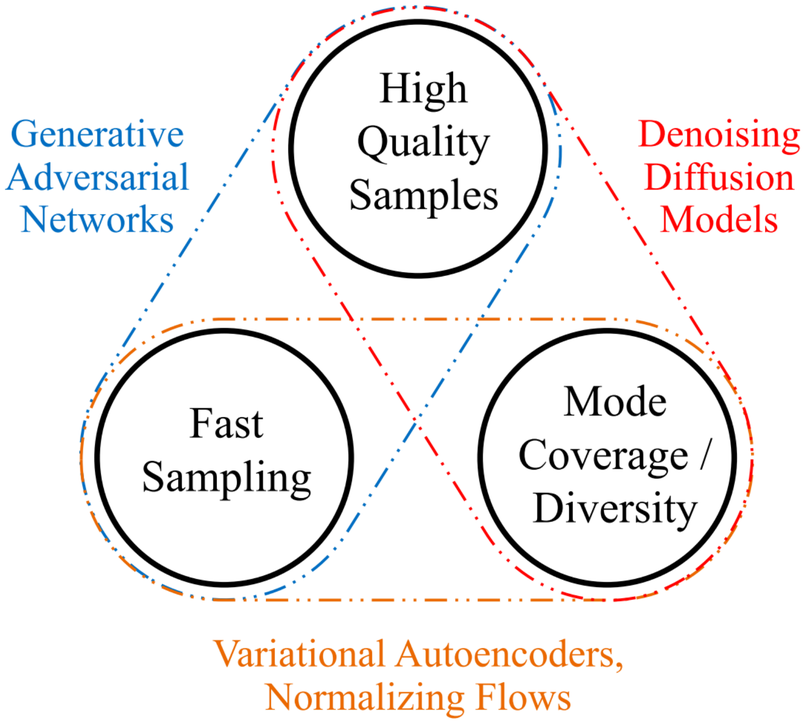




Diffusion Models

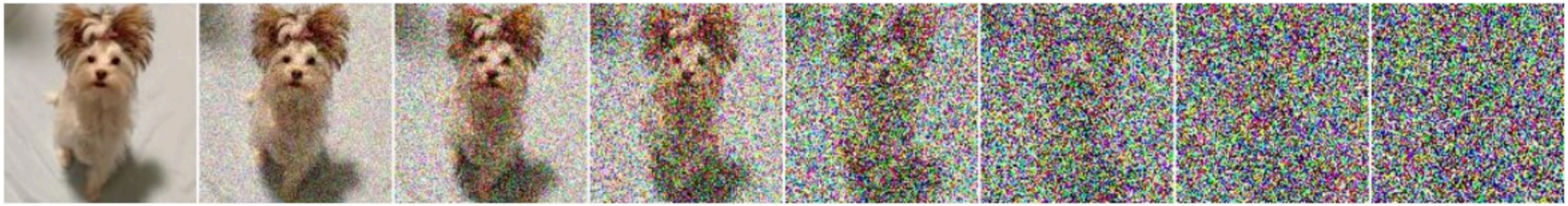
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Generative Trilemma



Tackling the Generative Learning Trilemma with Denoising Diffusion GANs

Diffusion: idea



$\mathbf{x}_0 \sim p_{\text{data}}$

$\mathbf{x}_T \sim p_{\text{noise}}$

Define the forward noising process and find its time-reversal

Score-Based Generative Modeling through Stochastic Differential Equations

Noising processes

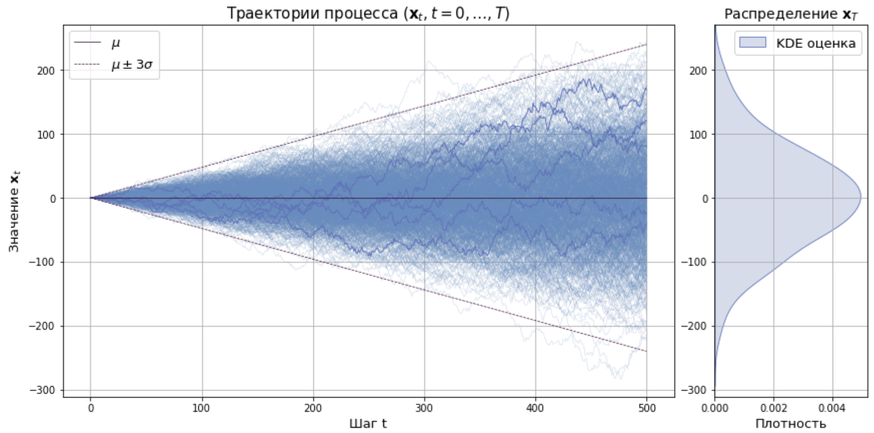
Variance Exploding (VE)

Variance Preserving (VP)

Markov process $q_{0,\dots,T}(\mathbf{x}_0, \dots, \mathbf{x}_T) = p_{\text{data}}(\mathbf{x}_0) \prod_{t=1}^{T} q_{t|t-1}(\mathbf{x}_t|\mathbf{x}_{t-1})$

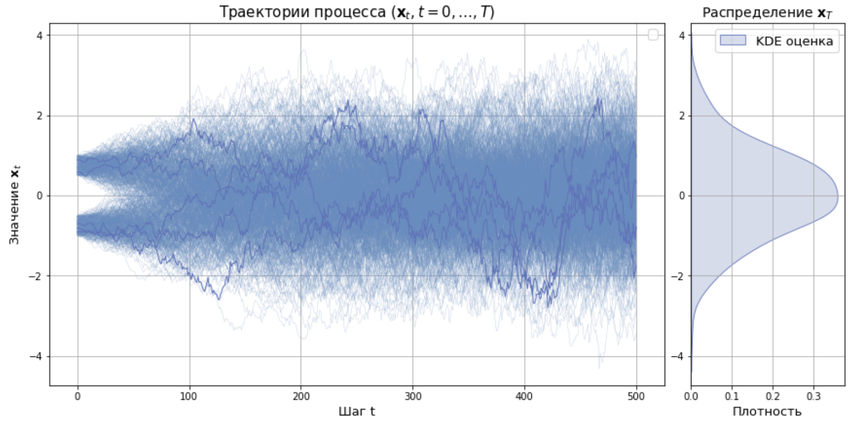
$$\mathbf{x}_t = \mathbf{x}_{t-1} + g(t)\boldsymbol{\varepsilon}_t \qquad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(0, I)$$

$$q_{t|t-1}(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t|\mathbf{x}_{t-1}, g^2(t)I)$$



$$\mathbf{x}_t = \sqrt{1-\beta_t}\,\mathbf{x}_{t-1} + \sqrt{\beta_t}\,\boldsymbol{\varepsilon}_t \qquad \boldsymbol{\varepsilon}_t \sim \mathcal{N}(0, I)$$

$$q_{t|t-1}(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t|\sqrt{1-\beta_t}\,\mathbf{x}_{t-1},\ \beta_t I)$$



$$q_{t|0}(\mathbf{x}_t|\mathbf{x}_0) = ?$$

$$\mathbf{x}_T \sim ?$$

Noising processes: sum of Gaussians

Variance Exploding (VE)

$$\mathbf{x}_t = \mathbf{x}_{t-1} + g(t)\boldsymbol{\varepsilon}_t$$

$$\mathbf{x}_t = \mathbf{x}_0 + \sum_{s=1}^{t} g(s)\boldsymbol{\varepsilon}_s$$

Variance Preserving (VP)

$$\mathbf{x}_t = \sqrt{1-\beta_t}\,\mathbf{x}_{t-1} + \sqrt{\beta_t}\,\boldsymbol{\varepsilon}_t$$

$$= \sqrt{1-\beta_t}\sqrt{1-\beta_{t-1}}\mathbf{x}_{t-2} + \sqrt{1-\beta_t}\sqrt{\beta_{t-1}}\boldsymbol{\varepsilon}_{t-1} + \sqrt{\beta_t}\boldsymbol{\varepsilon}_t = \dots$$

$$= \prod_{s=1}^{t}\sqrt{1-\beta_s}\cdot\mathbf{x}_0 + \sum_{s=1}^{t}\left(\prod_{\tau=s+1}^{t}\sqrt{1-\beta_\tau}\right)\sqrt{\beta_s}\cdot\boldsymbol{\varepsilon}_s =$$

$$= \sqrt{\alpha_t}\,\mathbf{x}_0 + \sum_{s=1}^{t}\frac{\sqrt{\alpha_t}}{\sqrt{\alpha_s}}\sqrt{\beta_s}\,\boldsymbol{\varepsilon}_s$$

$$\alpha_t = \prod_{s=1}^{t}(1-\beta_s)$$

Sum of independent Gaussian variables → Gaussian $q_{t|0}(\mathbf{x}_t|\mathbf{x}_0)$

Noising processes: statistics

Variance Exploding (VE)

$$\mathbf{x}_t = \mathbf{x}_{t-1} + g(t)\boldsymbol{\varepsilon}_t$$

$$\begin{aligned}\mathbb{E}_q[\mathbf{x}_t|\mathbf{x}_0] &= \mathbb{E}_q[\mathbf{x}_{t-1} + g(t)\boldsymbol{\varepsilon}_t|\mathbf{x}_0] = \\ &= \mathbb{E}_q[\mathbf{x}_{t-1}|\mathbf{x}_0] + g(t)\mathbb{E}_q[\boldsymbol{\varepsilon}_t|\mathbf{x}_0] \\ &= \mathbb{E}_q[\mathbf{x}_{t-1}|\mathbf{x}_0] = \ldots = \mathbb{E}_q[\mathbf{x}_0|\mathbf{x}_0] = \mathbf{x}_0\end{aligned}$$

Variance Preserving (VP)

$$\mathbf{x}_t = \sqrt{1-\beta_t}\,\mathbf{x}_{t-1} + \sqrt{\beta_t}\,\boldsymbol{\varepsilon}_t$$

$$\begin{aligned}\mathbb{E}_q[\mathbf{x}_t|\mathbf{x}_0] &= \mathbb{E}_q[\sqrt{1-\beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\varepsilon}_t|\mathbf{x}_0] = \\ &= \sqrt{1-\beta_t}\,\mathbb{E}_q[\mathbf{x}_{t-1}|\mathbf{x}_0] + \sqrt{\beta_t}\mathbb{E}_q[\boldsymbol{\varepsilon}_t|\mathbf{x}_0] \\ &= \sqrt{1-\beta_t}\,\mathbb{E}_q[\mathbf{x}_{t-1}|\mathbf{x}_0] = \ldots \\ &= \prod_{s=1}^{t}\sqrt{1-\beta_s}\cdot\mathbb{E}_q[\mathbf{x}_0|\mathbf{x}_0] = \sqrt{\alpha_t}\mathbf{x}_0\end{aligned}$$

Noising processes: statistics

Variance Exploding (VE)

$$\mathbf{x}_t = \mathbf{x}_{t-1} + g(t)\boldsymbol{\varepsilon}_t$$

$$\begin{aligned}
\mathrm{Var}_q[\mathbf{x}_t|\mathbf{x}_0] &= \mathrm{Var}_q[\mathbf{x}_{t-1} + g(t)\boldsymbol{\varepsilon}_t|\mathbf{x}_0] \\
&= \mathrm{Var}_q[\mathbf{x}_{t-1}|\mathbf{x}_0] + g^2(t)\mathrm{Var}_q[\boldsymbol{\varepsilon}_t|\mathbf{x}_0] = \\
&= \mathrm{Var}_q[\mathbf{x}_{t-1}|\mathbf{x}_0] + g^2(t)\cdot I = \ldots \\
&= \mathrm{Var}_q[\mathbf{x}_0|\mathbf{x}_0] + \sum_{s=1}^{t} g^2(s)\cdot I = \sum_{s=1}^{t} g^2(s)\cdot I
\end{aligned}$$

Variance Preserving (VP)

$$\mathbf{x}_t = \sqrt{1-\beta_t}\,\mathbf{x}_{t-1} + \sqrt{\beta_t}\,\boldsymbol{\varepsilon}_t$$

$$\begin{aligned}
\mathrm{Var}_q[\mathbf{x}_t|\mathbf{x}_0] &= \mathrm{Var}_q[\sqrt{1-\beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\boldsymbol{\varepsilon}_t|\mathbf{x}_0] = \\
&= (1-\beta_t)\mathrm{Var}_q[\mathbf{x}_{t-1}|\mathbf{x}_0] + \beta_t\mathrm{Var}_q[\boldsymbol{\varepsilon}_t|\mathbf{x}_0] = \\
&= (1-\beta_t)\mathrm{Var}_q[\mathbf{x}_{t-1}|\mathbf{x}_0] + \beta_t\cdot I
\end{aligned}$$

$$\begin{aligned}
I - \mathrm{Var}_q[\mathbf{x}_t|\mathbf{x}_0] &= I - \beta_t\cdot I - (1-\beta_t)\mathrm{Var}_q[\mathbf{x}_{t-1}|\mathbf{x}_0] = \\
&= (1-\beta_t)I - (1-\beta_t)\mathrm{Var}_q[\mathbf{x}_{t-1}|\mathbf{x}_0] = \\
&= (1-\beta_t)(I - \mathrm{Var}_q[\mathbf{x}_{t-1}|\mathbf{x}_0]) = \\
&= \prod_{s=1}^{t}(1-\beta_s)\cdot(I - \mathrm{Var}_q[\mathbf{x}_0|\mathbf{x}_0]) = \prod_{s=1}^{t}(1-\beta_s)\cdot I
\end{aligned}$$

$$\mathrm{Var}_q[\mathbf{x}_t|\mathbf{x}_0] = \left(1 - \prod_{s=1}^{t}(1-\beta_s)\right)\cdot I = (1-\alpha_t)\cdot I$$

Noising processes: transitions

Variance Exploding (VE)

$$\mathbf{x}_t = \mathbf{x}_{t-1} + g(t)\boldsymbol{\varepsilon}_t$$

$$q_{t|0}(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}\left(\mathbf{x}_t \mid \mathbf{x}_0, \sigma_t^2 \cdot I\right)$$

$$\sigma_t^2 = \sum_{s=1}^{t} g^2(s)$$

$$q_{T|0}(\mathbf{x}_T|\mathbf{x}_0) = \mathcal{N}\left(\mathbf{x}_T \mid \mathbf{x}_0, \sigma_T^2 \cdot I\right) \approx \mathcal{N}\left(\mathbf{x}_T \mid 0, \sigma_T^2 \cdot I\right)$$

$$q_T(\mathbf{x}_T) \approx \mathcal{N}\left(\mathbf{x}_T \mid 0, \sigma_T^2 \cdot I\right)$$

Variance Preserving (VP)

$$\mathbf{x}_t = \sqrt{1-\beta_t}\,\mathbf{x}_{t-1} + \sqrt{\beta_t}\,\boldsymbol{\varepsilon}_t$$

$$q_{t|0}(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}\left(\mathbf{x}_t \mid \sqrt{\alpha_t}\,\mathbf{x}_0, (1-\alpha_t) \cdot I\right)$$

$$\alpha_t = \prod_{s=1}^{t}(1-\beta_s) \to 0$$

$$q_{T|0}(\mathbf{x}_T|\mathbf{x}_0) = \mathcal{N}\left(\mathbf{x}_T \mid \sqrt{\alpha_T}\mathbf{x}_0, (1-\alpha_T) \cdot I\right) \approx \mathcal{N}\left(\mathbf{x}_T \mid 0, I\right)$$

$$q_T(\mathbf{x}_T) \approx \mathcal{N}\left(\mathbf{x}_T \mid 0, I\right)$$

Transition from data to pure independent noise

How to perform time reversal?

Sampling from time reversal:

$$\mathbf{x}_T \sim p_{\text{noise}}(\mathbf{x}_T) = \mathcal{N}(\mathbf{x}_T|0, I) \,/\, \mathcal{N}(\mathbf{x}_T|0, \sigma_T^2)$$
$$\mathbf{x}_{t-1} \sim q_{t-1|t}(\mathbf{x}_{t-1}|\mathbf{x}_t)$$

Calculating the backward transitions:

$$q_{t-1|t}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \frac{q_{t|t-1}(\mathbf{x}_t|\mathbf{x}_{t-1})\, q_{t-1}(\mathbf{x}_{t-1})}{q_t(\mathbf{x}_t)}$$

intractable

Modeling the time reversal

Replace true backward transition with trainable backward transition

$$p^{\theta}_{t-1|t}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1} \mid \mu^{\theta}_t(\mathbf{x}_t), \gamma_t^2 \cdot I)$$

Neural net

Trainable backward process:

$$p^{\theta}_{0,\dots,T}(\mathbf{x}_0, \dots, \mathbf{x}_T) = p_{\text{noise}}(\mathbf{x}_T) \prod_{s=1}^{T} p^{\theta}_{t-1|t}(\mathbf{x}_{t-1}|\mathbf{x}_t)$$

Ground truth forward noising process:

$$q_{0,\dots,T}(\mathbf{x}_0, \dots, \mathbf{x}_T) = p_{\text{data}}(\mathbf{x}_0) \prod_{t=1}^{T} q_{t|t-1}(\mathbf{x}_t|\mathbf{x}_{t-1})$$

Goal

$$p^{\theta}_{0,\dots,T}(\mathbf{x}_0, \dots, \mathbf{x}_T) = q_{0,\dots,T}(\mathbf{x}_0, \dots, \mathbf{x}_T)$$

DDPM training

Goal

$$p^{\theta}_{0,\dots,T}(\mathbf{x}_0, \dots, \mathbf{x}_T) = q_{0,\dots,T}(\mathbf{x}_0, \dots, \mathbf{x}_T)$$

Train via minimizing the KL divergence!

$$\mathrm{KL}\left(q_{0,\dots,T}(\mathbf{x}_0, \dots, \mathbf{x}_T) \,||\, p^{\theta}_{0,\dots,T}(\mathbf{x}_0, \dots, \mathbf{x}_T)\right) \to \min_{\theta}$$

DDPM training: decomposition

$$\mathrm{KL}\left(q_{0,\dots,T}(\mathbf{x}_0,\dots,\mathbf{x}_T)\,||\,p^{\theta}_{0,\dots,T}(\mathbf{x}_0,\dots,\mathbf{x}_T)\right) = \mathbb{E}_q \log \frac{q_{0,\dots,T}(\mathbf{x}_0,\dots,\mathbf{x}_T)}{p^{\theta}_{0,\dots,T}(\mathbf{x}_0,\dots,\mathbf{x}_T)} =$$

$$= \mathbb{E}_q \log \frac{p^{\text{data}}(\mathbf{x}_0) q_{T|0}(\mathbf{x}_T|\mathbf{x}_0) \prod_{t=2}^{T} q_{t-1|t,0}(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)}{p_{\text{noise}}(\mathbf{x}_T) p^{\theta}_{0|1}(\mathbf{x}_0|\mathbf{x}_1) \prod_{t=2}^{T} p^{\theta}_{t-1|t}(\mathbf{x}_{t-1}|\mathbf{x}_t)} \approx$$

const

small transition, ignore

$$\approx \text{const} + \mathbb{E}_q \log \frac{\prod_{t=2}^{T} q_{t-1|t,0}(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)}{\prod_{t=2}^{T} p^{\theta}_{t-1|t}(\mathbf{x}_{t-1}|\mathbf{x}_t)} = \text{const} + \sum_{t=2}^{T} \mathbb{E}_q \log \frac{q_{t-1|t,0}(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)}{p^{\theta}_{t-1|t}(\mathbf{x}_{t-1}|\mathbf{x}_t)} =$$

$$= \text{const} + \sum_{t=2}^{T} \mathbb{E}_{q_{0,t}(\mathbf{x}_0,\mathbf{x}_t)} \mathrm{KL}\left(q_{t-1|t,0}(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)\,||\,p^{\theta}_{t-1|t}(\mathbf{x}_{t-1}|\mathbf{x}_t)\right)$$

DDPM training: conditional backward transition

Equivalent objective

$$\sum_{t=2}^{T} \mathbb{E}_{q_{0,t}(\mathbf{x}_0, \mathbf{x}_t)} \mathrm{KL} \left(q_{t-1|t,0}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \,||\, p^{\theta}_{t-1|t}(\mathbf{x}_{t-1}|\mathbf{x}_t) \right) \to \min_{\theta}$$

??? Gaussian

Conditional backward transition:

$$q_{t-1|t,0}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \frac{q_{t|t-1}(\mathbf{x}_t|\mathbf{x}_{t-1}) q_{t-1|0}(\mathbf{x}_{t-1}|\mathbf{x}_0)}{q_{t|0}(\mathbf{x}_t|\mathbf{x}_0)}$$

All three are Gaussians → Gaussian!

In fact, for both VP and VE

$$q_{t-1|t,0}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}|A_t\mathbf{x}_t + B_t\mathbf{x}_0,\ C_t^2 \cdot I)$$

holds with some parameters A_t, B_t, C_t .

DDPM training: KL between Gaussians

$$\mathrm{KL}\left(\mathcal{N}(\mathbf{x}|\mu_1, \sigma_1^2 \cdot I) \;||\, \mathcal{N}(\mathbf{x}|\mu_2, \sigma_2^2 \cdot I) = \frac{1}{2}\left(d \cdot \frac{\sigma_1^2}{\sigma_2^2} + d \cdot \log \frac{\sigma_2^2}{\sigma_1^2} + \frac{1}{\sigma_2^2}\|\mu_1 - \mu_2\|^2 - d\right)\right.$$

Our case: conditional backward transition and model transition

$$q_{t-1|t,0}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}|A_t\mathbf{x}_t + B_t\mathbf{x}_0,\; C_t^2 \cdot I)$$

$$p^\theta_{t-1|t}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1} \mid \mu_t^\theta(\mathbf{x}_t), \gamma_t^2 \cdot I)$$

set $\gamma_t = C_t$ to minimize KL

The objective becomes equal to

$$\sum_{t=2}^{T} \mathbb{E}_{q_{0,t}(\mathbf{x}_0,\mathbf{x}_t)} \mathrm{KL}\left(q_{t-1|t,0}(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) \;||\; p^\theta_{t-1|t}(\mathbf{x}_{t-1}|\mathbf{x}_t)\right) =$$

$$= \sum_{t=2}^{T} \mathbb{E}_{q_{0,t}(\mathbf{x}_0,\mathbf{x}_t)} \frac{1}{2C_t^2} \left\|\mu_t^\theta(\mathbf{x}_t) - (A_t\mathbf{x}_t + B_t\mathbf{x}_0)\right\|^2 \to \min_\theta$$

DDPM training: denoiser form

Equivalent objective

$$\sum_{t=2}^{T} \mathbb{E}_{q_{0,t}(\mathbf{x}_0,\mathbf{x}_t)} \frac{1}{2C_t^2} \left\| \mu_t^\theta(\mathbf{x}_t) - (A_t\mathbf{x}_t + B_t\mathbf{x}_0) \right\|^2 \to \min_\theta$$

Parameterize neural net to mimic the target:

$$\mu_t^\theta(\mathbf{x}_t) = A_t\mathbf{x}_t + B_t D_t^\theta(\mathbf{x}_t)$$

The resulting objective

$$\sum_{t=2}^{T} \mathbb{E}_{q_{0,t}(\mathbf{x}_0,\mathbf{x}_t)} \frac{B_t^2}{2C_t^2} \left\| D_t^\theta(\mathbf{x}_t) - \mathbf{x}_0 \right\|^2 \to \min_\theta$$

independent objectives for different t
→ replace with any ω_t

$D_t^\theta(\mathbf{x}_t)$ tries to denoise $\mathbf{x}_t$ into clean $\mathbf{x}_0$

DDPM: training algorithm

Final form of the objective

$$\sum_{t=2}^{T} \omega_t \, \mathbb{E}_{q_{0,t}(\mathbf{x}_0, \mathbf{x}_t)} \| D_t^\theta(\mathbf{x}_t) - \mathbf{x}_0 \|^2 \to \min_\theta$$

Training algorithm

- Sample $\mathbf{x}_0$ from the dataset;
- Sample time step t;
- Sample standard Gaussian noise $\boldsymbol{\varepsilon}$;
- Perform data noising: $\mathbf{x}_t = \sqrt{\alpha_t}\,\mathbf{x}_0 + \sqrt{1-\alpha_t}\,\boldsymbol{\varepsilon}$ (VP) or $\mathbf{x}_t = \mathbf{x}_0 + \sigma_t\,\boldsymbol{\varepsilon}$ (VE);
- Make prediction $D_t^\theta(\mathbf{x}_t)$;
- Make gradient step on $\omega_t \| D_t^\theta(\mathbf{x}_t) - \mathbf{x}_0 \|^2$.

DDPM: sampling

General sampling algorithm:

- $\mathbf{x}_T \sim p_{\text{noise}}$
- $\mathbf{x}_{t-1} \sim p^{\theta}_{t-1|t}(\mathbf{x}_{t-1}|\mathbf{x}_t)$

Transition in our model:

$$p^{\theta}_{t-1|t}(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}|A_t\mathbf{x}_t + B_t D^{\theta}_t(\mathbf{x}_t), C_t^2 \cdot I)$$

Concrete algorithm:

- $\mathbf{x}_T \sim \mathcal{N}(0, I)$ (VP) or $\mathbf{x}_T \sim \mathcal{N}(0, \sigma_T^2 I)$ (VE)
- $\mathbf{x}_{t-1} = A_t\mathbf{x}_t + B_t D^{\theta}_t(\mathbf{x}_t) + C_t\boldsymbol{\varepsilon}_t$

Iterative denoising:
combine current noisy sample with denoiser prediction

DDPM: summary

Training

General

$$\mathrm{KL}\left(q_{0,\dots,T}(\mathbf{x}_0,\dots,\mathbf{x}_T)\,||\,p^{\theta}_{0,\dots,T}(\mathbf{x}_0,\dots,\mathbf{x}_T)\right)\to\min_{\theta}$$

Model-specific

$$\sum_{t=2}^{T}\omega_t\,\mathbb{E}_{q_{0,t}(\mathbf{x}_0,\mathbf{x}_t)}\|D^{\theta}_t(\mathbf{x}_t)-\mathbf{x}_0\|^2\to\min_{\theta}$$

Idea: $D^{\theta}_t(\mathbf{x}_t)$ tries to denoise $\mathbf{x}_t$ into clean $\mathbf{x}_0$

Sampling

General

$$\mathbf{x}_T\sim p_{\text{noise}}$$
$$\mathbf{x}_{t-1}\sim p^{\theta}_{t-1|t}(\mathbf{x}_{t-1}|\mathbf{x}_t)$$

Model-specific

$$\mathbf{x}_T\sim\mathcal{N}(0,I)\text{ (VP) or }\mathbf{x}_T\sim\mathcal{N}(0,\sigma_T^2 I)\text{ (VE)}$$
$$\mathbf{x}_{t-1}=A_t\mathbf{x}_t+B_tD^{\theta}_t(\mathbf{x}_t)+C_t\boldsymbol{\varepsilon}_t$$

Idea: combine current noisy sample with prediction of the clean image

What next?

1. How to perform conditional generation?
2. Ways to control quality and diversity of generated samples?